var x = 5, y = 2;

document.write(x % y);

A) 2.5 B) 1 C) 2 D) 0

2. What is the output?

let a = "5";

let b = 5;

console.log(a == b, a === b);

A) true true B) false false C) true false D) false true

3. What is the output?

javascript

var x = 10;

var y = "2";

console.log(x + y);

A) 12 B) "102" C) 102 D) NaN

4. What is the output?

var x = 10;

var y = "2";

console.log(x - y);

A) 8 B) "102" C) 102 D) NaN

5. What is the output?

javascript

let x = 3;

let y = x++ + ++x;

console.log(y);

A) 6 B) 7 C) 8 D) 9

6. What is the output?

javascript

console.log(parseInt("42px"));

A) 42 B) "42" C) NaN D) 42px

7. What is the output?

javascript

```
console.log(parseFloat("3.14abc"));
```

A) 3 B) 3.14 C) NaN D) "3.14"

8. What is the output?

javascript

```
console.log(isNaN("Hello"));
```

A) true B) false C) NaN D) "Hello"

9. What is the output?

javascript

```
var a = 5;
if (a = 3) {
  console.log("Yes");
} else {
  console.log("No");
}
```

A) Yes B) No C) Error D) undefined

10. What is the output?

javascript

```
var i = 0;
while (i < 3) {
  console.log(i);
  i++;
}
```

A) 1 2 3 B) 0 1 2 C) 0 1 2 3 D) Infinite loop

11. What is the output?

javascript

```
var count = 0;
do {
```

```
  console.log(count);
  count++;
} while (count < 0);
```

A) Nothing B) 0 C) 1 D) Infinite loop

12. What is the output?

```
javascript
for (var i = 0; i < 5; i++) {
  if (i === 2) continue;
  console.log(i);
}
```

A) 0 1 2 3 4 B) 0 1 3 4 C) 0 1 2 D) 0 1 2 3

13. What is the output?

```
javascript
var x = 10;
switch (x) {
  case 10: console.log("A");
  case 20: console.log("B"); break;
  default: console.log("C");
}
```

A) A B) A B C) A B C D) B

14. What is the output?

```
javascript
var a = 5;
var b = a++ + a;
console.log(b);
```

A) 10 B) 11 C) 12 D) 13

15. What is the output?

```
javascript
console.log(typeof null);
```

A) "null" B) "object" C) "undefined" D) "number"

16. What is the output?

javascript

```
console.log(2 + 3 + "4");
```

A) "234" B) "54" C) 9 D) "9"

17. What is the output?

javascript

```
console.log("2" + 3 + 4);
```

A) "234" B) "54" C) 9 D) NaN

18. What is the output?

javascript

```
let x = 5;
let y = x > 3 ? "Big" : "Small";
console.log(y);
```

A) Big B) Small C) true D) 5

19. What is the output?

javascript

```
var sum = 0;
for (var i = 1; i <= 4; i++) {
  sum += i;
}
console.log(sum);
```

A) 4 B) 10 C) 15 D) 20

20. What is the output?

javascript

```
function test() {
  var x = 10;
}
test();
```

console.log(x);

A) 10 B) undefined C) ReferenceError D) null

SECTION B: ARRAY METHODS (Questions 21–38)

21. What is the output?

javascript

var arr = [1, 2, 3];

arr.push(4);

console.log(arr.length);

A) 3 B) 4 C) 5 D) 10

22. What is the output?

javascript

var arr = [1, 2, 3, 4];

var x = arr.pop();

console.log(arr, x);

A) [1,2,3] 4 B) [2,3,4] 1 C) [1,2,3,4] 4 D) [1,2,3] undefined

23. What is the output?

javascript

var arr = [1, 2, 3];

var x = arr.shift();

console.log(arr, x);

A) [1,2,3] 1 B) [2,3] 1 C) [1,2] 3 D) [2,3] 3

24. What is the output?

javascript

var arr = [1, 2, 3];

arr.unshift(0);

console.log(arr);

A) [1,2,3,0] B) [0,1,2,3] C) [1,2,3] D) [0,1,2]

25. What is the output?

javascript

```
var arr = [10, 5, 100, 1];
arr.sort();
console.log(arr[0]);
```

A) 1 B) 5 C) 10 D) 100

26. What is the output?

javascript

```
var arr = [10, 5, 100, 1];
arr.sort(function(a, b){ return a - b; });
console.log(arr);
```

A) [1,5,10,100] B) [10,100,5,1] C) [1,10,100,5] D) [100,10,5,1]

27. What is the output?

javascript

```
var arr = [1, 2, 3];
arr.reverse();
console.log(arr);
```

A) [1,2,3] B) [3,2,1] C) [2,1,3] D) [1,3,2]

28. What is the output?

javascript

```
var a = [1, 2];
var b = [3, 4];
console.log(a.concat(b).length);
```

A) 2 B) 3 C) 4 D) 6

29. What is the output?

javascript

```
var arr = ["a", "b", "c"];
console.log(arr.indexOf("b"));
```

A) 0 B) 1 C) 2 D) -1

30. What is the output?

javascript

```
var arr = ["a", "b", "c"];
console.log(arr.indexOf("z"));
```

A) 0 B) 2 C) -1 D) undefined

31. What is the output?

javascript

```
var arr = [1, 2, 3, 4];
var result = arr.slice(1, 3);
console.log(result);
```

A) [1,2] B) [2,3] C) [2,3,4] D) [1,2,3]

32. What is the output?

javascript

```
var arr = [1, 2, 3, 4];
arr.splice(1, 2);
console.log(arr);
```

A) [1,4] B) [1,2,3,4] C) [2,3] D) [1,2]

33. What is the output?

javascript

```
var arr = [1, 2, 3];
console.log(arr.join("-"));
```

A) "123" B) "1-2-3" C) "1,2,3" D) [1,2,3]

34. What is the output?

javascript

```
var nums = [1, 2, 3, 4];
var evens = nums.filter(function(n){ return n % 2 === 0; });
console.log(evens);
```

A) [1,3] B) [2,4] C) [1,2,3,4] D) [4]

35. What is the output?

javascript

```
var nums = [1, 2, 3];
var doubled = nums.map(function(n){ return n * 2; });
console.log(doubled);
```

A) [1,2,3] B) [2,4,6] C) [1,2,3,1,2,3] D) 6

36. What is the output?

javascript

```
var arr = [1, 2, 3];
console.log(arr.toString());
```

A) "1,2,3" B) "123" C) [1,2,3] D) "1 2 3"

37. What is the output?

javascript

```
var arr = [5, 6, 7, 8];
var x = arr.length;
arr[10] = 99;
console.log(arr.length);
```

A) 4 B) 8 C) 11 D) 99

38. What is the output?

javascript

```
var arr = ["red", "green", "blue"];
var x = arr.includes("green");
console.log(x);
```

A) 1 B) 2 C) true D) false

SECTION C: DOM (Questions 39–50)

39. What is the output?

html

```
<p id="demo">Hello</p>
<script>
  document.getElementById("demo").textContent = "World";
```

</script>

A) Hello B) World C) Hello World D) Nothing

40. Given A B , what does document.querySelectorAll(".tip").length return?
A) 0 B) 1 C) 2 D) undefined

41. Given <h1 id="title">Hi</h1>, which selects the element?
A) document.querySelector("h1") B) document.getElementById("#title") C) Both A and a corrected getElementById("title") D) document.getElementsByTagName(".title")

42. What is the output?

html

<script>

const box = document.querySelector("#box");

box.innerHTML = "Hi";

console.log(box.textContent);

</script>

A) "Hi" B) Hi C) strong D) undefined

43. What does this do?

javascript

const btn = document.querySelector("#btn");

btn.addEventListener("click", () => {

btn.textContent = "Clicked!";

});

A) Changes the button text when clicked B) Changes the button text immediately on page load C) Removes the button D) Nothing happens

44. What is the output?

html

<ul id="list">
A

<script>

const list = document.querySelector("#list");

```
  const item = document.createElement("li");
  item.textContent = "B";
  list.appendChild(item);
  console.log(list.children.length);
</script>
```

A) 1 B) 2 C) 3 D) 0

45. Which method removes an element from the DOM?
A) element.delete() B) element.remove() C) element.hide() D) element.clear()

46. Given , what does card.classList.add("active") do?
A) Replaces the class "card" B) Adds "active" to the class list C) Removes "card" D) Creates a new div

47. What is the output?

```
html
<input id="name" value="John">
<script>
  const input = document.querySelector("#name");
  console.log(input.value);
</script>
```

A) undefined B) null C) "John" D) <input>

48. In event.target, what does target refer to?
A) The event type B) The element that triggered the event C) The parent document D) The browser window

49. What does event.preventDefault() do in a form submit handler?
A) Stops the event from bubbling B) Stops the form from reloading/submitting normally C) Deletes the form D) Clears all inputs

50. What is the output?

```
html
<p id="p">Text</p>
<script>
  const p = document.getElementById("p");
  p.style.color = "blue";
</script>
```

A) Text turns blue B) Text is deleted C) Nothing changes D) The page reloads

---

ANSWER KEY

Section A: Code Output

Q	Answer	Explanation
1	**B**	5 % 2 = 1 (remainder)
2	**C**	== compares value only (true), === compares value and type (false)
3	**B**	String + number → concatenation: "10" + "2" → "102"
4	**A**	- forces numeric conversion: 10 - 2 = 8
5	**C**	x++ returns 3 (then x=4), ++x makes x=5 and returns 5 → 3 + 5 = 8
6	**A**	parseInt stops at non-numeric → 42
7	**B**	parseFloat stops at non-numeric → 3.14
8	**A**	"Hello" is not a number → isNaN returns true
9	**A**	a = 3 is assignment (truthy), so "Yes" prints
10	**B**	Prints 0, 1, 2
11	**B**	do...while runs at least once → prints 0
12	**B**	continue skips 2 → 0 1 3 4
13	**B**	No break after case 10, so falls through → A then B

Q	Answer	Explanation
14	**B**	a++ returns 5 (a=6), then a=6 → 5 + 6 = 11
15	**B**	Historical JS quirk: typeof null is "object"
16	**B**	2+3=5, then 5 + "4" → "54"
17	**A**	"2"+3 → "23", "23"+4 → "234"
18	**A**	5 > 3 is true → "Big"
19	**B**	1+2+3+4 = 10
20	**C**	x is local to test(), so console.log(x) outside → ReferenceError

Section B: Array Methods

Q	Answer	Explanation
21	**B**	push adds 1 element → length 4
22	**A**	pop removes last (4), array becomes [1,2,3]
23	**B**	shift removes first (1), array becomes [2,3]
24	**B**	unshift adds to front → [0,1,2,3]
25	**A**	Default sort() is alphabetical: "1","10","100","5" → first is 1
26	**A**	Numeric comparator a-b sorts ascending → [1,5,10,100]
27	**B**	reverse → [3,2,1]

Q	Answer	Explanation
28	**C**	concat joins → length 4
29	**B**	"b" is at index 1 (zero-based)
30	**C**	Not found → -1
31	**B**	slice(1,3) extracts index 1 up to (not including) 3 → [2,3]
32	**A**	splice(1,2) removes 2 elements starting at index 1 → [1,4]
33	**B**	join("-") → "1-2-3"
34	**B**	filter keeps evens → [2,4]
35	**B**	map doubles → [2,4,6]
36	**A**	toString → "1,2,3"
37	**C**	Setting index 10 extends length to 11
38	**C**	includes returns boolean → true

Section C: DOM

Q	Answer	Explanation
39	**B**	textContent replaces text → "World"
40	**C**	Two .tip elements → length 2
41	**C**	querySelector("h1") works; getElementById("title") works (without #)

Q	Answer	Explanation
42	**B**	textContent of Hi is "Hi"
43	**A**	Listener fires on click → changes text
44	**B**	One existing + one appended → 2 children
45	**B**	remove() removes the element
46	**B**	classList.add adds "active"
47	**C**	.value reads input value → "John"
48	**B**	target is the element that triggered the event
49	**B**	Stops default form submission behavior
50	**A**	.style.color = "blue" turns text blue